

**RETAIL DATA ENGINEERING PIPELINE: TECHNICAL
IMPLEMENTATION AND BUSINESS INSIGHTS REPORT**

Project Framework and Technical Documentation

Prepared For: ABC Retail Solutions Stakeholders

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Date: 1 Jun 2026

Version: 1.0

Revision History

Version	Date	Author	Description
1.0	2026-06-01	Muhammed Zachariya	Initial Document Release

Executive Summary

This report details the successful implementation of an end-to-end data engineering pipeline for ABC Retail Solutions. Faced with challenges including duplicate records, inconsistent naming conventions, and sensitive PII exposure, this project established a robust framework for automated data ingestion, cleaning, and transformation. Key achievements include the resolution of critical data quality issues and the implementation of PII masking to ensure governance compliance. Insights generated from the curated datasets reveal that Electronics and Laptops drive the highest revenue shares, providing a reliable business intelligence foundation for strategic decision-making.

Table of Contents

TABLE OF CONTENTS

Executive Summary 1

1. Introduction 2

1.1 Project Background 2

1.2 Business Problem Statement 3

1.3 Project Objectives 3

2. Dataset Description 4

2.1 Product Details Dataset 4

2.2 Retail Transaction Dataset 5

3. Solution Architecture 6

3.1 High-Level Architecture 6

3.2 Data Flow Architecture 7

3.3 Data Model Diagram.....8

4. Data Ingestion 8

5. Data Cleaning and Transformation 10

5.1 Missing Value Handling 10

5.2 Data Standardization 11

5.3 Date Format Standardization 12

5.4 Duplicate Handling 12

6. Data Governance and PII Protection	13
7. KPI Generation and Business Metrics	15
8. Curated Dataset Creation	17
9. Power BI Dashboard Development	18
9.1 Executive Summary Dashboard	18
9.2 Product Performance Dashboard	19
9.3 Regional Insights Dashboard	20
9.4 Payment Analysis Dashboard	21
9.5 Data Quality and Governance Dashboard	22
10. Business Insights	23
11. Assumptions	24
12. Future Enhancements	25
13. Conclusion	26
Appendix	27

1. Introduction

1.1 Introduction

ABC Retail Solutions is a multinational retail company operating through both online and offline sales channels. The company collects transactional sales data from multiple source systems. However, inconsistencies such as duplicate records, missing values, inconsistent product naming conventions, invalid quantities, inconsistent category values, varying date formats, and exposure of Personally Identifiable Information (PII) create challenges in generating accurate business insights.

1.2 Project Objectives

- Ingest retail transaction data from multiple source files.
- Perform data quality validation checks and handle missing values.
- Standardize product and category information across datasets.
- Protect sensitive customer information through PII masking and governance.
- Generate business KPIs and create curated datasets for reporting.
- Build an interactive Power BI dashboard for business stakeholders.

2. Solution Architecture and Technology Stack

2.1 Architecture Overview

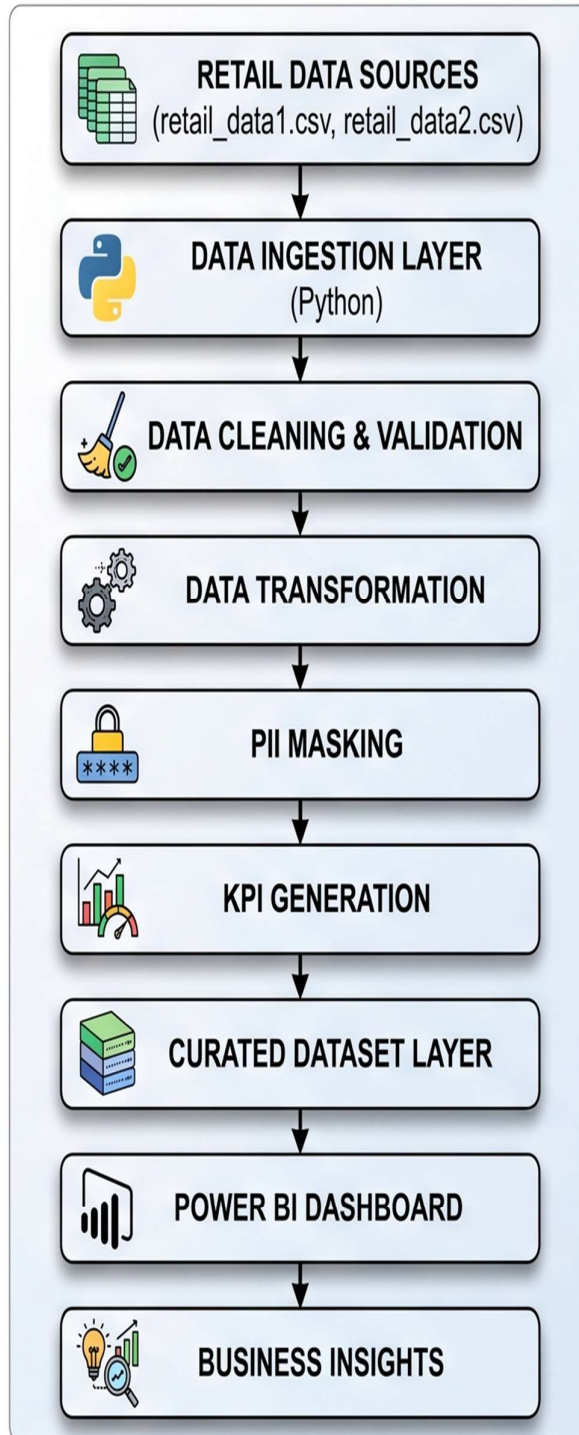


Figure 3.1: High-Level Data Engineering Pipeline Architecture

2.2 Data Flow Workflow

1. Source Ingestion: Retail CSV files loaded via Python Pandas.
2. Data Validation and Cleaning: Removal of duplicates and invalid records.
3. Standardization: Aligning naming conventions and formats.
4. Security: Implementation of PII Masking.
5. Aggregation: Business KPI generation and Curated Data export.

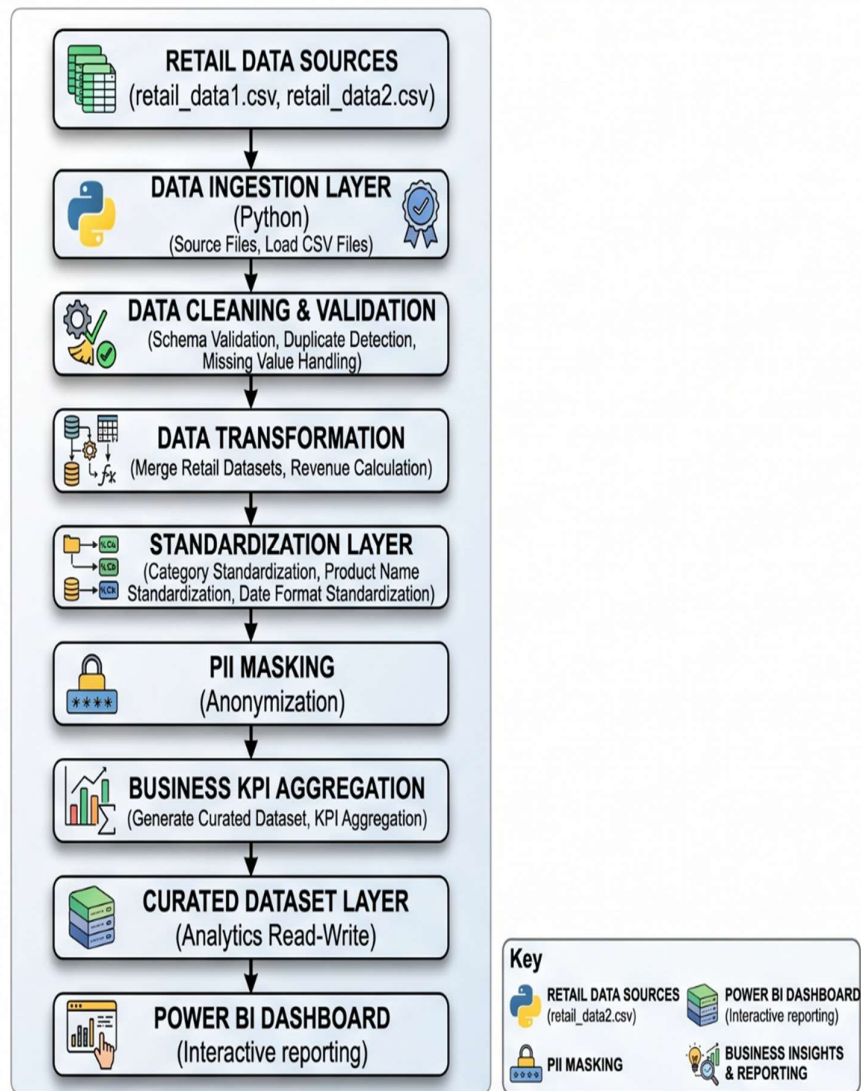


Figure 3.2: Detailed Data Flow Diagram

Figure 3.2: Sequential Data Processing Workflow.

2.3 Data Model diagram

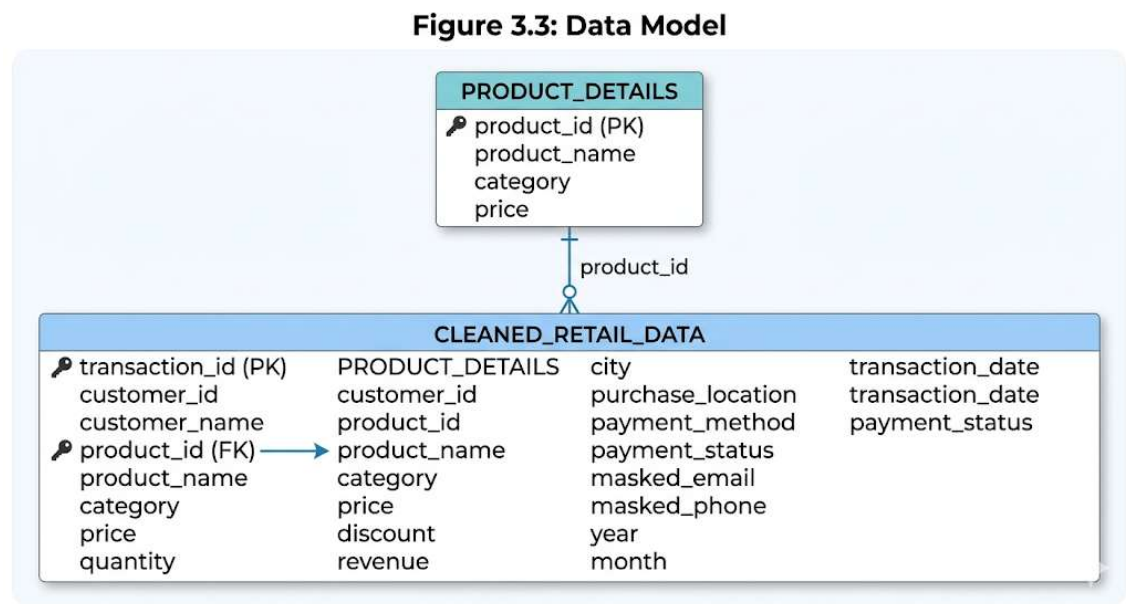


Figure 3.3: Data Model Diagram

2.3 Technology Stack

Table 3.1: Project Technology and Tools

Category	Tool / Library
Programming Language	Python (Pandas, NumPy)
Storage Format	CSV, Parquet
Visualization	Power BI Desktop
Version Control	GitHub

3. Dataset Description

3.1 Product Details Dataset

This dataset contains master reference information for products.

- **Columns:** product_id, product_name, category, price.
- **Purpose:** Product validation, category validation, missing value imputation, and data enrichment.

3.2 Retail Transaction Dataset

The retail transaction dataset consists of records collected from two independent source systems.

- **Columns:** transaction_id, customer_id, customer_name, product_id, price, product_name, category, purchase_location, city, transaction_date, quantity, payment_method, discount, email, phone, payment_status.

Source File	Record Count
retail_data1	4,243
retail_data2	4,251
Total Records	8,494

4. Data Ingestion

Data ingestion was performed using Python and Pandas to ensure a robust starting point for the pipeline.

1. Load retail_data1.csv and retail_data2.csv.

2. Load product_details.csv.
3. Validate schema consistency across sources.
4. Merge both retail datasets into a single unified dataframe.

5. Data Cleaning and Transformation

5.1 Duplicate and Missing Value Handling

- **Duplicate Handling:** Duplicate records were identified and removed using Pandas detection functionality.
- **Missing Values:** Missing values were populated using product master reference data for fields including product_name, category, and price.

5.2 Standardization Logic

Field	Inconsistency Addressed	Standardized Format
Product Name	TV, Tv, tv	TV
Category	ELEC, FURN, CLOTH, HOME	Electronics, Furniture, Clothing, Home Appliances
Date	DD/MM/YYYY, MM-DD-YYYY	Standardized Datetime Format

Validation

Records with invalid or non-positive quantities were removed from the final dataset to maintain data integrity.

6. PII Protection and Data Governance

To comply with data governance requirements, Personally Identifiable Information (PII) was protected through masking techniques.

- **Email Masking:** Original: johnsmith@gmail.com → Masked: j***@gmail.com
- **Phone Number Masking:** Original: 9876543210 → Masked: *****3210

7. KPI Calculations

7.1 Total Revenue

Revenue was calculated using the following formula:

$\text{Revenue} = \text{Price} \times \text{Quantity} \times (1 - \text{Discount})$

7.2 Additional Metrics

- Total Orders & Total Customers.
- Average Order Value (AOV).
- Revenue broken down by Category, City, Product, and Payment Method.

8. Curated Dataset Creation

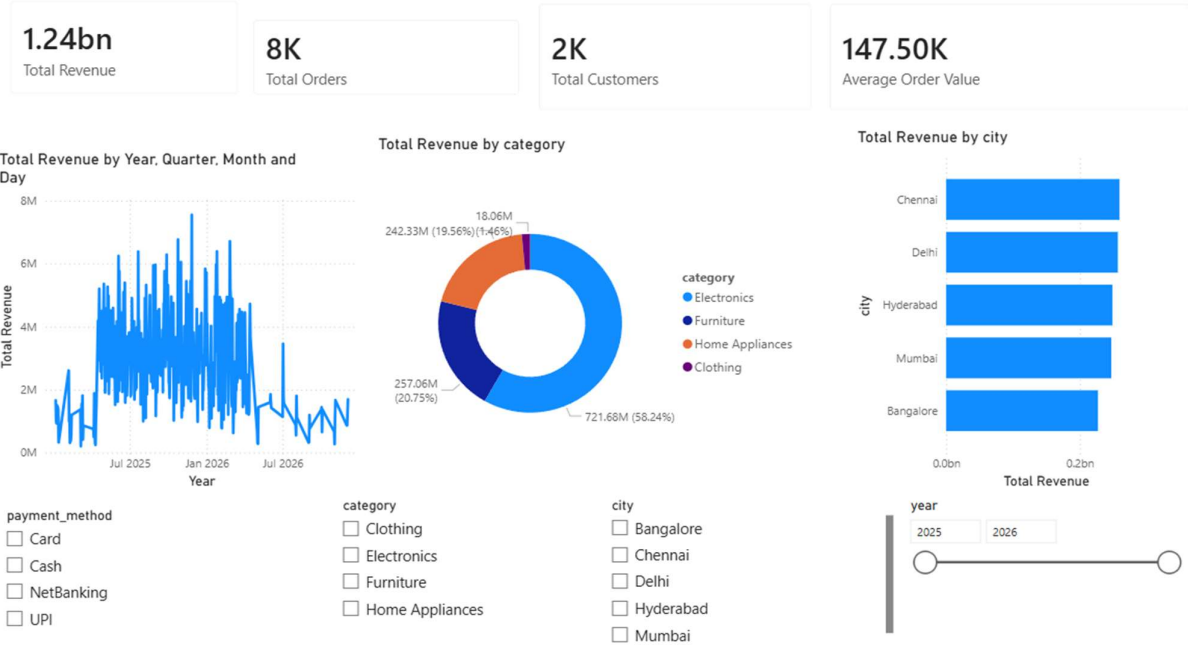
The transformation process concludes with the generation of curated files optimized for reporting:

- cleaned_retail_data.csv / cleaned_retail_data.parquet
- revenue_by_category.csv
- revenue_by_city.csv
- top_products.csv

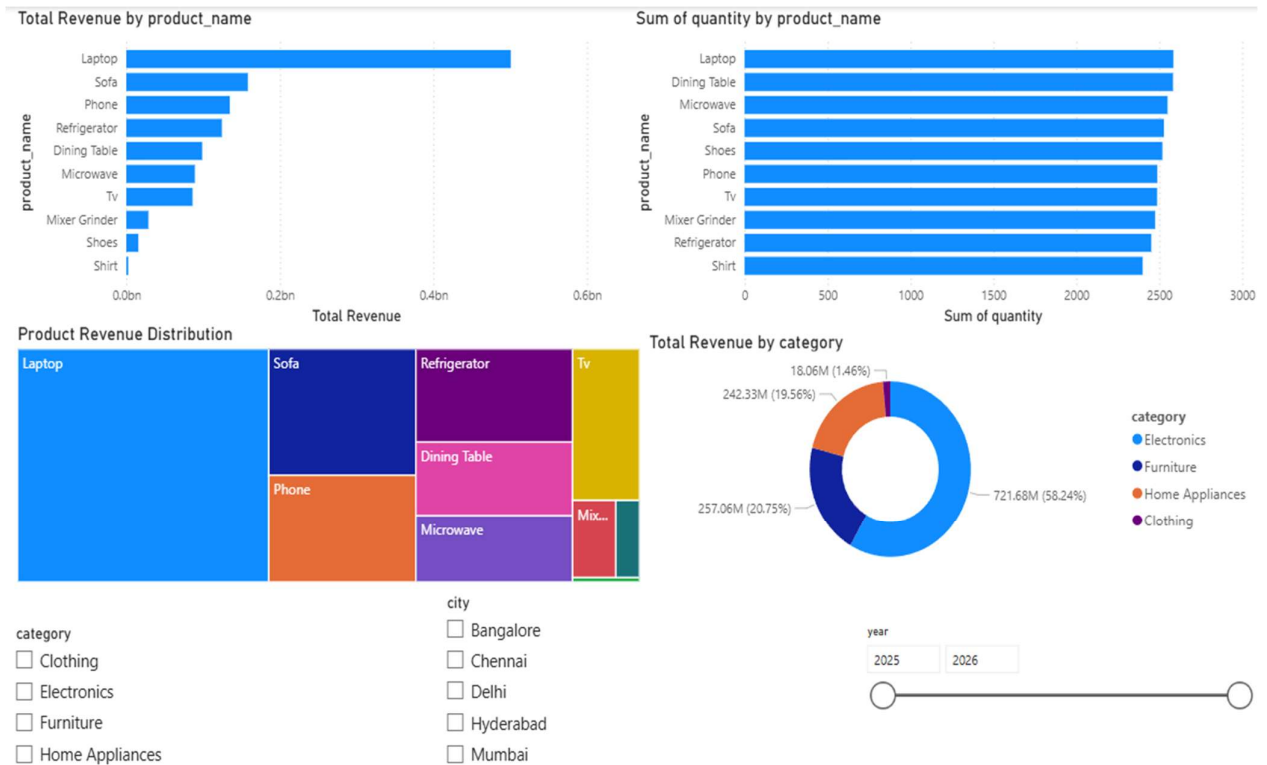
- data_quality_report.csv

9. Power BI Dashboard Development

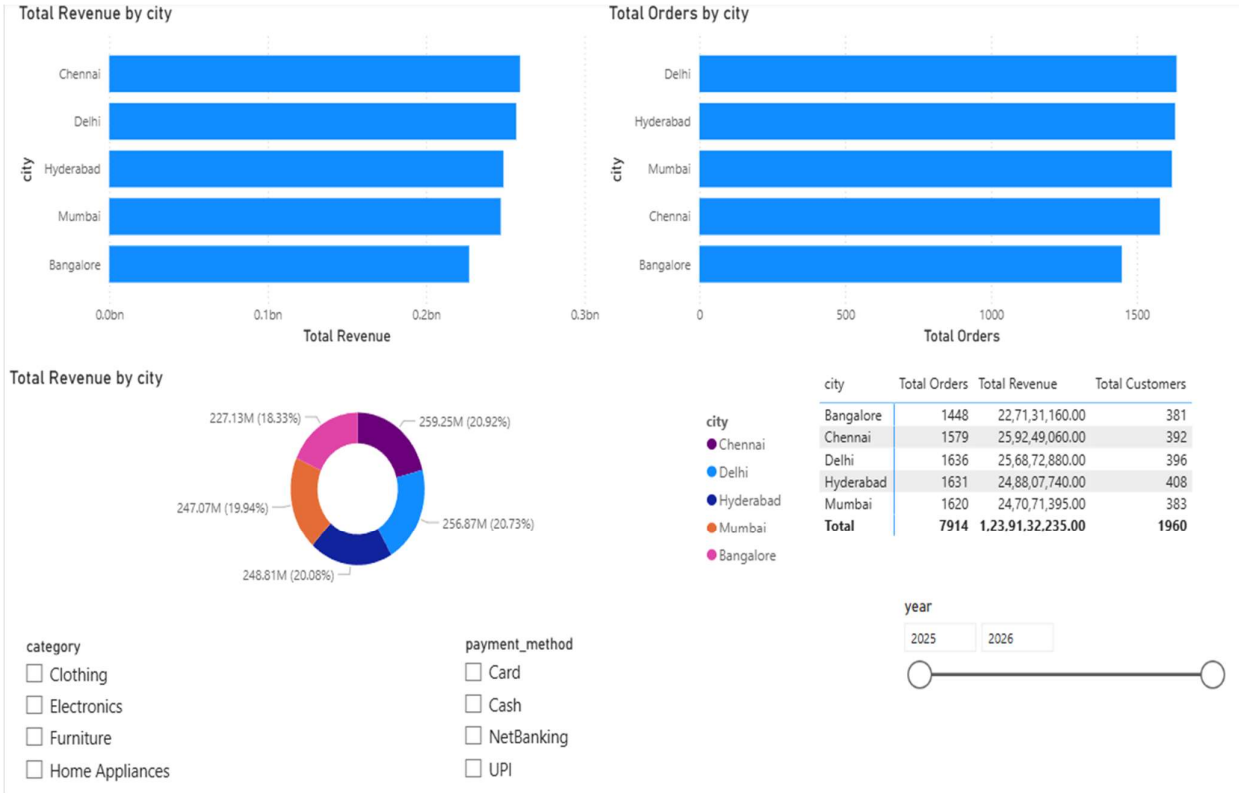
9.1 Executive Summary Dashboard



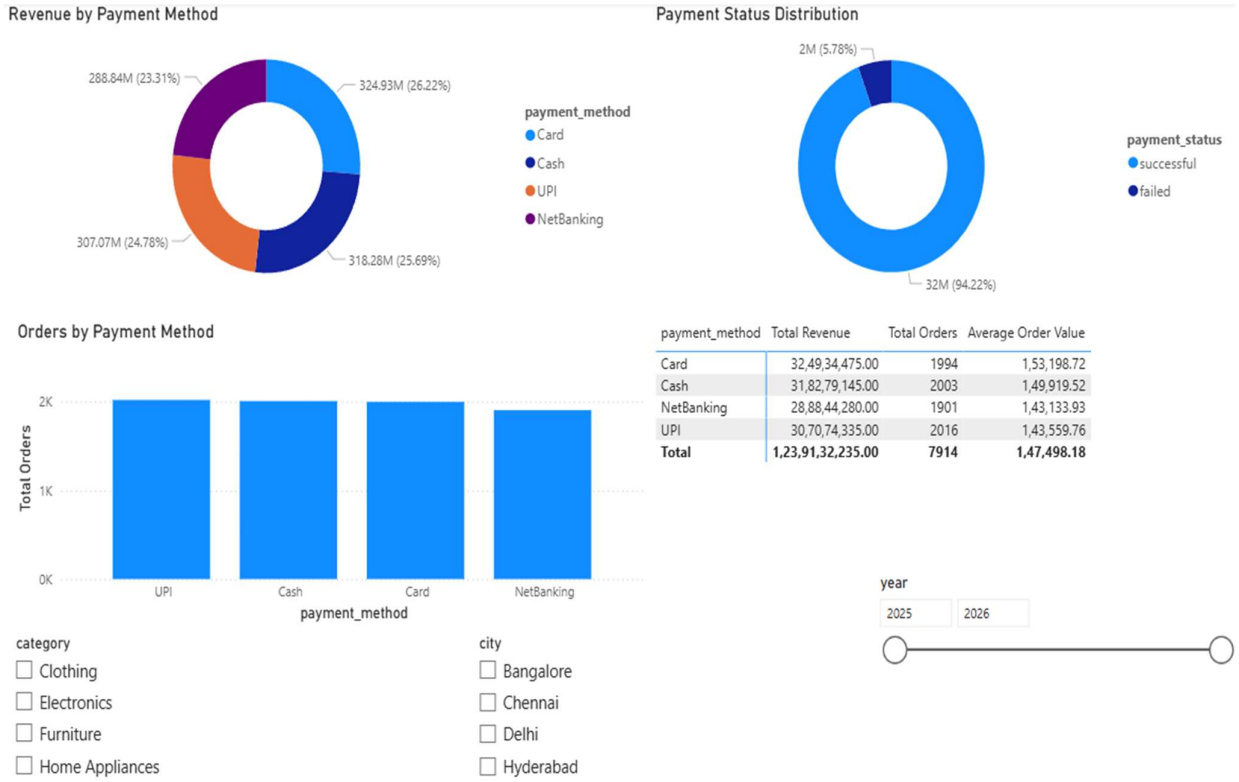
9.2 Product Performance Dashboard



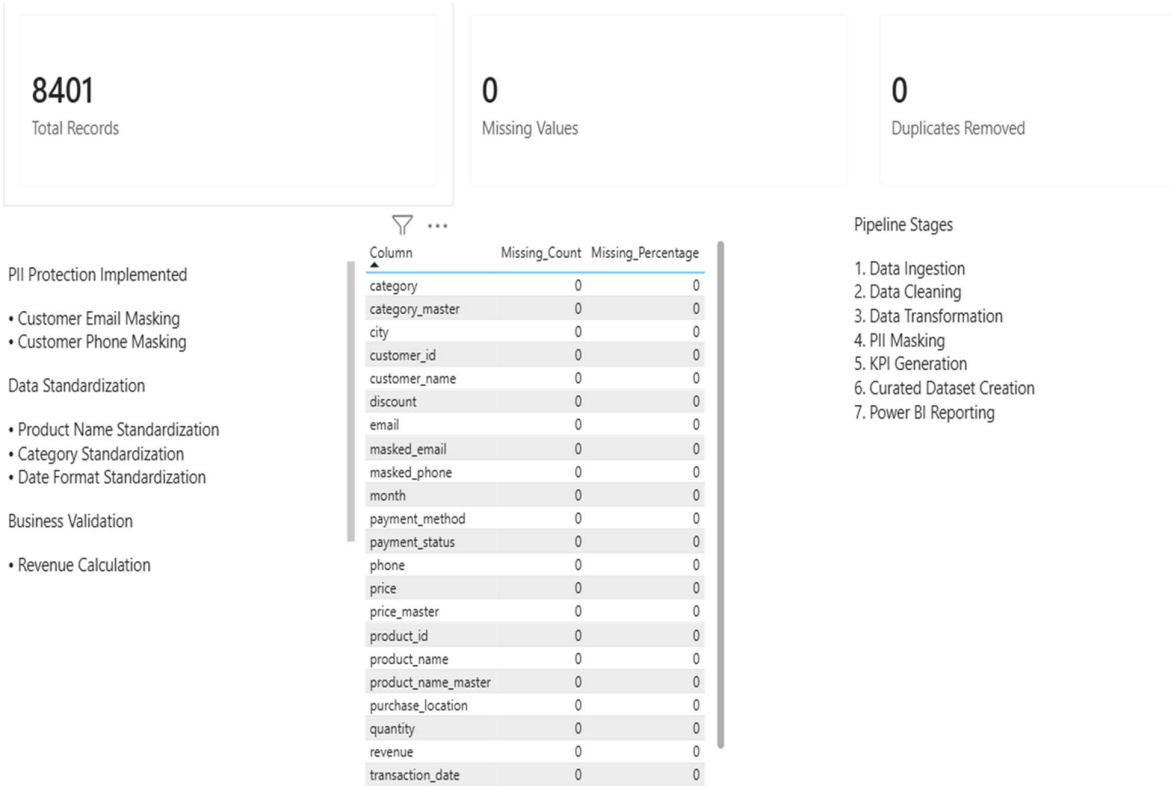
9.3 Regional Insights Dashboard



9.4 Payment Analysis Dashboard



9.5 Data Quality and Governance Dashboard



A multi-page Power BI dashboard was developed to provide comprehensive visibility.

Page	Key Visuals
1. Executive Summary	Total Revenue, Total Orders, Revenue Trends, Revenue by Category & City.
2. Product Performance	Top 10 Products (Revenue/Quantity), Product Treemap, Category Analysis.
3. Regional Insights	Revenue & Orders by City, Regional KPI Tables, Share by City.

Page	Key Visuals
4. Payment Analysis	Revenue by Payment Method, Payment Status Distribution, KPI Table.
5. Quality & Governance	Total Records, Missing Values, Duplicates Removed, Pipeline Workflow.

10. Key Business Insights

- **Electronics** contributed the highest share of revenue.
- **Laptops** generated the highest individual product revenue.
- Revenue distribution across cities was relatively balanced.
- **UPI, Card, Cash, and Net Banking** all contributed significantly to sales volume.
- More than **90%** of transactions were successfully completed.

11. Assumptions

- Missing product information can be accurately derived from the product master data.
- The product master data is the absolute "Single Source of Truth."
- Invalid quantities are considered data entry errors and are excluded from reporting.
- Masked customer information is sufficient for demographic reporting without exposing identities.

12. Future Enhancements

- Integration with **Azure Data Factory** for orchestration.
- Migration to **Azure Data Lake** for scalable storage.
- Implementation of **Incremental Data Loading** and real-time streaming.
- **Predictive Sales Forecasting** using machine learning models.

13. Conclusion

This project successfully implemented an end-to-end retail data engineering pipeline. By resolving data quality issues and securing sensitive information, the solution provides a reliable foundation for data-driven decision-making through interactive Power BI reporting.